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A Low cost area-efficient modified Russian peasant multiplier(MRPM) for biomedical applications

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ABSTRACT

The essentiality of efficient hardware multipliers has increased as they have a huge impact on the performance and the power consumption of digital systems. This work proposes a novel 8-bit unsigned multiplier architecture which combines the Modified Russian Peasant algorithm with a Kogge-Stone adder. The aim of the proposed design is to minimize resource usage and power consumption while retaining high speed and accuracy. The effective optimization of the multiplier design led to a significant reduction in area utilization and overall power consumption, making it highly suitable for implementing in resource constrained systems. The design is validated using Xilinx Vivado tool and is implemented on Nexys A7 FPGA board. The results show that the proposed multiplier utilizes fewer devices (50%–70% less) and has comparative speed and power consumption with respect to the conventional multipliers. The proposed multiplier design can be used for biomedical systems such as wearable health monitors and implantable medical devices, where compact size and energy efficiency are key considerations. This efficient multiplier architecture ensures prolonged battery life and reliable performance, enabling advanced signal processing tasks such as signal filtering, enhancement of medical imaging and noise reduction in hearing aids. These features emphasize the multiplier's potential to enable substantial enhancements in high-performance, area-sensitive, and power-sensitive biomedical applications, which will contribute to the forthcoming compact and energy-efficient healthcare systems.

1. Introduction

The design and realization of efficient multipliers is of great importance within the study area of digital systems and signal processing [1] as they play a vital role in the continuous development of these systems. Multipliers are the most time consuming and resource intensive components in these digital systems and they have a huge impact on determining the overall effectiveness and power [2] consumption of these computer-based units. One particularly important area in which energy-efficient, compact, and high-performance computational hardware are needed is the development of wearable health monitors and implantable biomedical devices [3]. These devices need innovations that find the right balance between computational accuracy, resource efficiency, and lower power consumption to ensure longer battery life, accuracy and reliability.

Although, traditional multiplier architectures, such as the Array Multiplier [4], Booth Multiplier [4] etc., have been extensively researched and optimized over the years, their efficiency in computing applications is often limited by high resource utilization, high power consumption, and increased design complexity. These problems become more pronounced for battery-operated and resource-constrained systems like biomedical devices, where every microampere of current and square millimetre of silicon matter. This signifies the strong demand for novel multiplier architectures that can address such issues, and provide a balance between computational accuracy, resource efficiency, and power consumption.

Recent advancements in VLSI technology have led to the design and development of variety of multipliers with an objective of minimizing power consumption and improving computational speed. For example, the Modified Booth Multiplier [5] and Wallace Tree Multiplier [6] have shown great improvements on speed and area efficiency. Many multipliers have emerged as a solution to achieve reduced area [7] and power consumption by sacrificing computational accuracy, such solutions are promising for error-tolerant applications, they are not always suitable in all compute methods like they are not suitable

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for accuracy-critical biomedical systems, where precision is essential. Thus, despite these advancements in the area of digital system design, there remains a need for further optimization, especially for biomedical applications [2] where power and space constraints are stringent. Therefore, novel multiplier designs that can overcome these constraints without compromising performance are highly needed.

The Russian Peasant Multiplication (RPM) algorithm is famous for its simplicity and efficiency for developing optimized multiplier architectures. It is based on iterative shift-and-add operations and has attracted attention due to its potential merits of reduced power and resource usage. Previously, approaches that leverage algorithmic optimization have been explored. For instance, researchers have implemented multipliers based on shift-and-add techniques to simplify hardware operations. But existing implementations do not provide the necessary speed and parallelism to satisfy the performance demands of modern applications. This algorithm provides better performance, without penalizing the design with resource utilization, when coupled with Kogge-Stone adder which is built on parallel prefix addition technique, thus maximizing the speed of operations without introducing timing issues. Hence, the Modified RPM algorithm, combined with the Kogge-Stone adder, delivers a promising solution for cost and energy efficient systems. This combination leverages the simplicity and efficiency of the RPM algorithm with the fast, parallel prefix operation of the Kogge-Stone adder. The integration of such high-speed adders with algorithmically optimized multipliers can help to bridge the trade-off between area savings and power savings with performance without sacrificing accuracy.

This work proposes a novel 8-bit unsigned multiplier architecture based on the Modified RPM algorithm integrated with Kogge-Stone adder. This design aims to overcome the drawbacks of conventional multipliers by significantly reducing resource usage and power consumption while maintaining high computational speed and accuracy. The Xilinx Vivado tool is used to evaluate the proposed multiplier and is implemented on Nexys A7 FPGA development board. Using the parallel prefix operation of the Kogge-Stone adder and the simplicity and efficiency of the Russian Peasant algorithm, this design provides a promising solution for biomedical devices with energy constraints, such as wearable health monitors and implantable medical devices. Through the results, it is shown that the optimized RPM has achieved substantial reductions with respect to Look-Up Table (LUT) usage and power consumption and the results also give evidence of the ability of this design to improve the performance and energy efficiency of biomedical systems, there by contributing to the improvement of healthcare technology by enhancing their functionality, reliability.

The rest of the paper is structured as follows: Section 2 provides a detailed review of related literature analysing the gaps of conventional multiplier architectures. Section 3 describes the proposed Modified RPM algorithm with the Kogge-Stone adder technique. The tools used and implementation methodology is discussed in Section 4. Section 5 presents the results obtained and thoroughly analyses the performance metrics such as resource utilization, power consumption and computational efficiency. Finally, Section 6 concludes the work with insights on the design's suitability for biomedical applications and future optimization perspectives.

2. Related works

Vedic multipliers were shown to be appropriate for modern platforms like the Nexys 7 board, where lower computational complexity was obtained. Yogita Bansal, Charu Madhu and Pardeep Kaur [8] presented a low-power shift-and-add multiplier design for the embedded-systems environment, with the aim of decreasing switching activity and operand gating. By contrast, the Vedic multipliers emphasize a parallel approach and speed at the expense of larger hardware size and high power consumption, which makes them less preferred for biomedical applications constrained in resources. E. Jagadeeswara Rao, T.Ramanjaneyulu and K. Jayaram Kumar [9] suggested the use of a Modified Russian Peasant Multiplier with a Han-Carlson Adder to reduce the delay in partial product addition. The design achieved more than 80% speed improvement when compared to RPMs using RCA, CSA, and 8:2 compressors.

In another research study, design and FPGA implementation of 4×4 Vedic Multiplier using different architectures, Samiksha Dhole, Sayali Shembalkar, Tirupati Yadav, and Prasheel Thakre [10] stated that ripple carry, carry-lookahead, and carry-save adders were used to implement Vedic multipliers. Penchalaiah and Siva Kumar VG [11] proposed a Kogge-Stone Adder (KSA) architecture that outperforms the Carry Skip Adder (CSKA) by providing higher speed, lower power consumption, and reduced area for 8 to 64-bit additions. Their work addresses the limitations of CSKA in high-speed applications through an efficient Parallel Prefix Adder design. Raju B. R. [12] presented an efficient multiplier architecture involving the Kogge-Stone Adder for speeding up operation and slowing down the arithmetic circuits. It proves that for high-speed applications, integrating the KSA with the multiplier design shall give utmost performance compared to the conventional adders. In similar work high-speed reconfigurable FIR filter implementing RPM with Sklansky adder by K. Gunasekaran, M. Manikandan [13] optimized RPM by substituting the wallace adder in the system with introducing a carry-select with a sklansky adder. This change improves the performance of reconfigurable FIR filters by showing less latency, power consumption, and area. In VLSI implementation of an improved multiplier for FFT Computation in biomedical applications, the authors Arathi Ajay and R. Mary Lourde [3] proposed an optimal FFT computing for EEG analysis. The study also investigates several multipliers such as Booth, Wallace Tree, and Booth-encoded Wallace Tree multipliers and their effect on performance. It refines EEG signal classification for biomedical application by inserting Booth-encoded Wallace Tree Multipliers for fast FFT calculations. In order to improve FFT-based signal processing, this work shows the importance of efficient multiplier architectures.

Pedro Paz and Mario Garrido [14] proposed complex multiplier architectures with DSP48E1/E2 slices for high-speed FPGA-based applications. While they are worthy for FFT and signal processing, these methods cannot be applied to resource-constrained systems like biomedical implants due to their dependence on dedicated DSP blocks. Instead, the MRPM architecture uses general-purpose logic and Kogge-Stone adders, where full accuracy is achieved with less area and power, making it suitable for biomedical applications where conserving DSP slices and efficiency go simultaneously. Khurshid [15] proposed an approximate multiplier for error-tolerant applications, such as in image processing. In biomedical systems, any minor deviation can affect clinical reliability. In recent work [16], the multi-level approximate multiplier employed modified full adders to minimize power (up to 27.7%), area (up to 22.9%), and gate count (up to 22.8%) for error-resilient applications such as image processing. However, the error tolerance was up to 5%, which is unsuitable for biomedical systems that require strict precision. On the other hand, our MRPM design guarantees complete accuracy in computations using general-purpose FPGA logic and Kogge-Stone adders, providing a reliable and efficient alternative for applications where accuracy is critical, while still maintaining area and power efficiency. .

The existing multiplier designs highlights different designs, each addressing different trade-offs related to speed, power consumption, and area utilization. Vedic multipliers are known for their efficiency and simplicity, while other methods like Wallace and Booth multipliers offer specific advantages in different contexts. The performance of the RPM algorithm has been improved significantly by introduction of different type of adder compressors and many different types of adders; few other traditional algorithms were significantly improved by fast parallel prefix adders like Kogge-Stone adder. However, the integration of advanced adders with resource-efficient multiplier designs has not been explored sufficiently. Hence, there persists a challenge

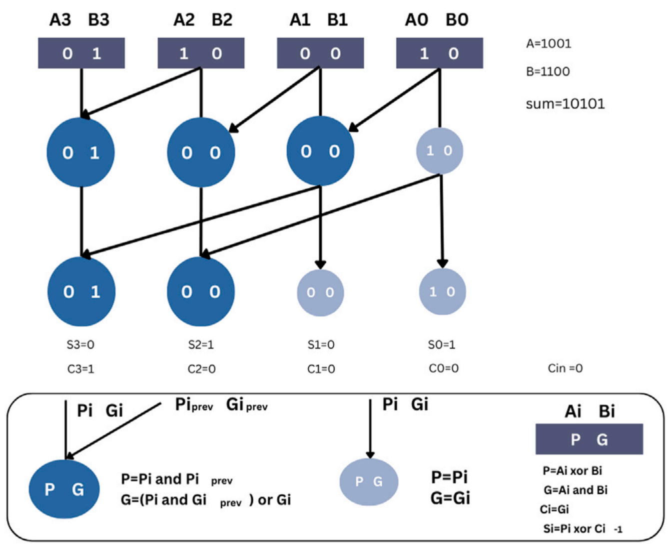

Fig. 1. Kogge-Stone adder.

in developing a solution that simultaneously optimizes power, area, and computational speed tailored to the resource constrained environments such as biomedical applications. The proposed Modified Russian Peasant Multiplier (MRPM) addresses these issues by providing an efficient, low-power, and compact architecture suitable for biomedical applications [2] like wearable and implantable medical devices.

3. Proposed Russian peasant multiplier

The RPM [13] algorithm is an efficient technique that performs multiplication by halving and doubling numbers. Every iteration of the process involves doubling one number (by left-shifting) and continuously halving the other number (by right-shifting). The value of the doubled operand is added to the result if the halved operand is odd. If not, it is neglected. This technique efficiently reduces the multiplication operation, making it simpler to compute and implement in hardware as bit shifting and repeated addition is used instead of storing. The primary disadvantage of the traditional RPM particularly for operands with a greater bit width is the accumulation of partial products which introduces some latency. Due to their sequential structure, traditional adders like ripple carry adders, which limit the multiplier's overall speed, further increase this delay during the accumulation phase.

To overcome the delay while keeping the computational efficiency, the Kogge-Stone Adder (KSA) [17] has been adopted to the RPM algorithm. KSA is popularly recognized for its parallel prefix architecture that can contribute to high speed and highly scalable carry propagation in wide bit-widths. Unlike ripple or carry-lookahead adders which degrade in performance with scaling, KSA has a constant logarithmic delay, which is quite suitable for the iterative and delay-sensitive MRPM structure.Its high-speed features, along with balanced hardware utilization, are a good trade-off between speed and area for biomedical applications with resource and power constraints.

A recognized parallel prefix adder for its rapid addition speed and low latency is the Kogge-Stone adder. Using a tree-like structure, the Kogge-Stone adder computes carry propagation in parallel, in contrast with traditional adders that propagate carries sequentially. The pre-processing, carry generation, and post-processing phases create this structure. In preprocessing stage, generate(G) and propagate(P) are computed for each bit of the input operands, defined as $Gi = (Ai.Bi)\, and\, Pi = (Ai \oplus Bi)$. The carry propagation time is significantly reduced by the carry generation stage, which uses a hierarchical prefix tree to compute carry signals for every bit location in parallel. Also, the sum is computed using the propagate values and the carry signals in the post-processing step as shown in Fig. 1. The Kogge-Stone adder is particularly well-suited for high-speed arithmetic operations in hardware implementations due to its parallel nature.

Therefore, in the modified design integrating the Russian multiplier with the Kogge-Stone adder leverages the area efficiency of the Russian multiplier and the speed of the Kogge-Stone adder creating a multiplier that is both compact and fast. The algorithm of MRPM is given in Algorithm 1.

3.1. Algorithm

Algorithm 1 Modified Russian Multiplier Algorithm using Kogge-Stone Adder

Require: 8-bit multiplicand A and 8-bit multiplier B
Ensure: 16-bit product P

Initialize A (multiplicand) and B (multiplier)
Set a 16-bit result $P \leftarrow 0$
Convert A and B to binary representation if necessary
while $B \neq 1$ **do**
  Left shift B and right shift A
  if B is odd **then**
    $P \leftarrow P + A$
  end if
end while
Integrate Kogge-Stone Adder for parallel prefix computation
Introduce Carry Propagation (P) and Generate (G) terms
for each bit position i **do**
  Compute initial propagate and generate: $P_i = A_i \oplus B_i, \quad G_i = A_i \cdot B_i$
  Compute carry propagation: $P = P_i \wedge P_{prev}, \quad G = (P_i \cdot G_{prev}) + G_i$
  Compute final sum and carry: $C_i = G_i, \quad S_i = P_i \oplus C_{i-1}$
end for
Reduce partial products using Kogge-Stone Adder and align result
Obtain 16-bit product P

Algorithm detailing the proposed architecture of the MRPM, which is designed to efficiently perform 8-bit unsigned multiplication with minimal power consumption and area requirements, making it highly suitable for applications like biomedical devices. The architecture takes two inputs, A and B (8-bit each), and operates iteratively. Initially, the logic checks if A is greater than zero, ensuring the multiplication process continues as long as required. A right barrel shifter divides A by 2 (logical right shift) in each iteration, while a left barrel shifter doubles B (logical left shift). A critical component is the logical unit that checks if A is odd; when true, B is added to the running total using a high-speed Kogge-Stone adder. Control logic is used to terminate the condition by checking if A has become less than or equal to 1. These iterative additions and shifts are combined to generate the final product of A and B. This design is optimized for compactness and energy efficiency. While barrel shifters handle shifting operations efficiently the Kogge-Stone adder guarantees quick computation and minimal delay. Feedback loops between the shifters, the logical units, and the adder are used in order to smooth the iterative processing.

The block diagram of the proposed MRPM is shown in Fig. 2 where the first operand (A) is iteratively checked for oddness to determine whether the second operand (B) should be added to the result. A is halved and B is doubled in each iteration. The condition "check if A is odd" is intentional and follows the standard RPM procedure for multiplication using shift-and-add operations and later the addition operation is carried out using Kogge-Stone adder. Also the architecture diagram of RPM and Kogge-Stone adder is shown in Fig. 3.

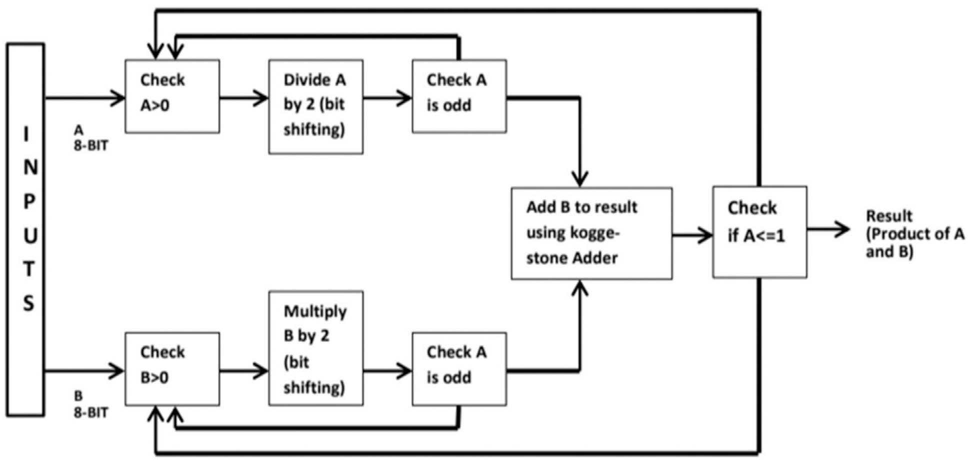

Fig. 2. Block diagram of 8-bit MRPM.

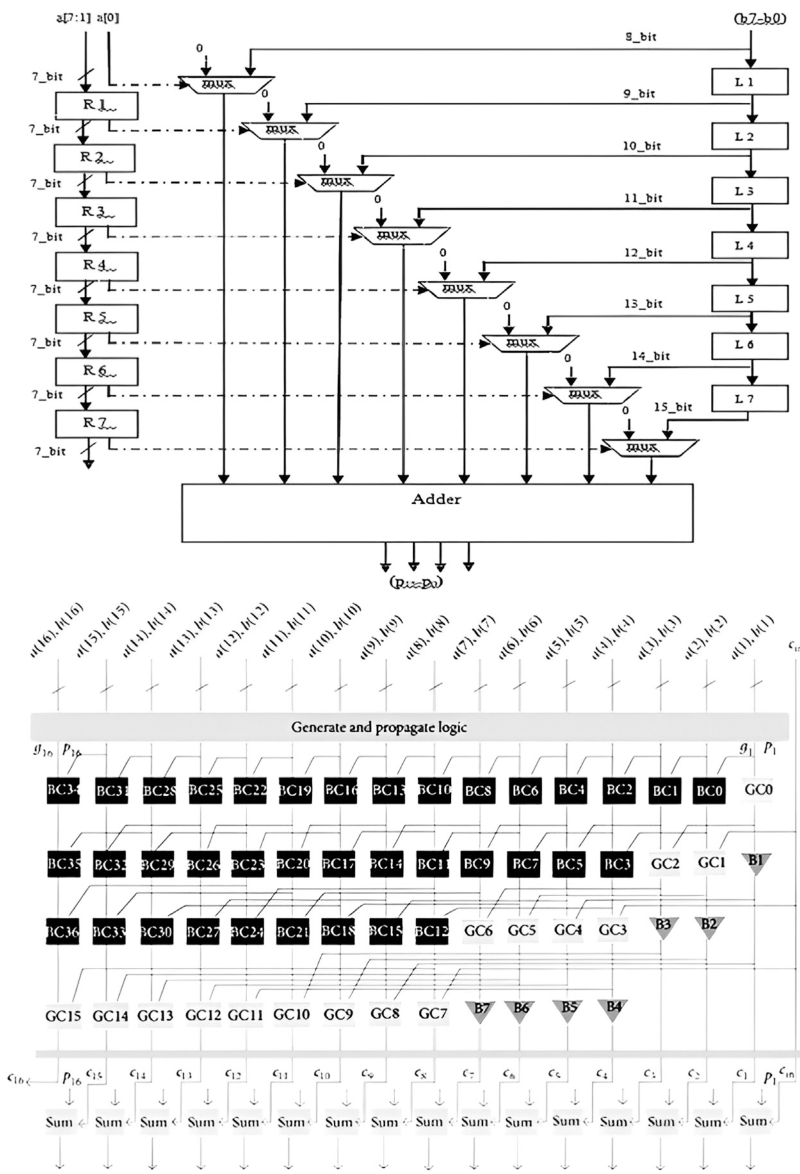

Fig. 3. Architecture diagram of the proposed 8-bit MRPM. The upper section illustrates the iterative shift-and-add logic, while the lower section shows the Kogge-Stone adder's generate-propagate logic for high-speed parallel addition.

4. Hardware implementation

The proposed MRPM is designed and thoroughly tested with the Xilinx Vivado design suite in order to perform FPGA-based simulation, synthesis and hardware deployment. The verilog coded architecture consists of the multiplier that is a modified RPM algorithm using Kogge-Stone adder individual components could be developed and validated more quickly by using this modular design approach. A testbench was employed to verify operations involved like iterative shifting, odd-checking, and controlled addition under various test cases.

The proposed 8-bit MRPM architecture was implemented on the Nexys A7 FPGA development board. Input operands `a_in` and `b_in` are given via the on-board slide switches, while the 16-bit multiplication result is displayed using the on-board LEDs, ranging from `result[15]` (MSB) to `result[0]` (LSB). The hardware setup is illustrated in Fig. 4.

The operational accuracy was completely verified through a very comprehensive testbench simulation involving scenarios for best case, worst case, and some random test vectors. The proposed multiplier is essentially an exact multiplier, which implies that it will give valid output for any testcase.

After functional validation the design was synthesized to produce a register-transfer-level (RTL) representation that was best suited for FPGA implementation. The synthesized netlist consists of key components such as a right barrel shifter for halving input A, a left barrel shifter for doubling input B, and Kogge-Stone adder for efficient addition. Control logic was implemented to ensure efficient operation of the multiplier and supervise the iterative process.

By creating a bit-stream file, the synthesized design was then dumped onto the Nexys A7 FPGA board, which is equipped with the Artix-7 XC7A100T FPGA. The hardware is then successful deployed with the designs of operational accuracy and stability were confirmed through real-time testing.

The hardware implementation process has demonstrated the efficiency and adaptability of the proposed multiplier, so that it can be used in systems that require small and effective arithmetic units. This implementation shows the application in embedded systems and biomedical application where hardware utilization and energy efficiency are crucial.

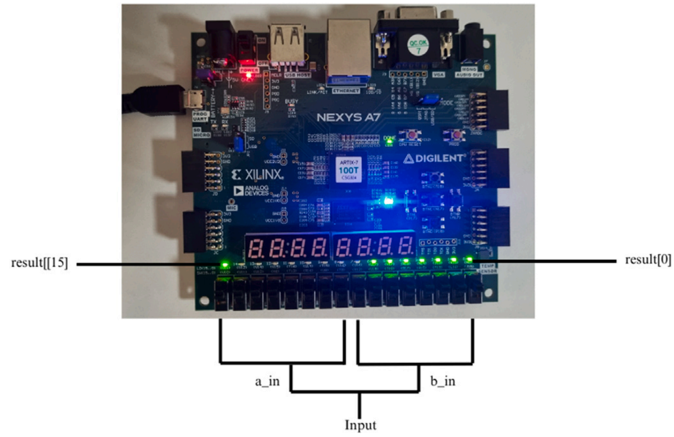

Fig. 4. Hardware implementation of the proposed 8-bit MRPM on the Nexys A7 FPGA board.

Table 1
Performance metrics.

Performance metrics	Values
Maximum clock frequency	0.714 MHz
Slice LUTs (63 400)	68
Slice Registers (126 800)	58
Slice (15 850)	24
LUT as logic (63 400)	68
Bonded IOB (210)	36
BUFGCTRL (32)	1
Nets	215
Leaf cells	196
Dynamic power	1.136 W
Delay	17.217 ns

5. Results and discussion

The simulation result of the proposed multiplier was verified by Vivado 2024.1 as shown in Fig. 5. The input numbers $a_{in}[7:0]$ and $b_{in}[7:0]$ are two 8-bit operands that need to be multiplied, go signal is to start the process and clk signal is for the clock pulses. The output of the product is a 16 bit number, result[15:0] and rdy signal indicates if the system have finished work or not. Once operation is completed and ready for the next test case, the ready signal becomes high.

Table 1 presents a detailed overview of the performance metrics related to resource utilization and power consumption for the proposed multiplier architecture. The design achieves a maximum clock frequency of 0.714 MHz, defining the upper limit of its operating speed. In terms of logic resources, it uses 68 Slice LUTs out of 63,400 available, and 58 Slice Registers out of 126,800, indicating a minimal use of FPGA logic and sequential elements. Additionally, only 24 slices are occupied out of a possible 15,850, further showcasing the design's compactness. For logic-specific resources, 68 LUTs are used purely as logic, reflecting efficient combinational logic implementation. Input/Output resource usage is also modest, with 36 Bonded IOBs out of 210 used for external interfacing. The clocking resource usage is minimal as well, with just 1 BUFGCTRL out of 32, suggesting low complexity in the clock distribution network. The design involves 215 nets, which represent the total signal routing paths, and utilizes 196 leaf cells, indicating the basic standard cells instantiated in the design. Power-wise, the architecture consumes only 1.136 W of dynamic power, which is suitable for energy-conscious applications. Finally, the design has a critical path delay of 17.217 ns, which determines the latency for the longest data path through the architecture. Collectively, these results demonstrate that the proposed multiplier is both area and power-efficient, making it an excellent choice for low-power digital systems.

The proposed multiplier shows significant benefits over a number of recognized architectures, including the Vedic, Array, Booth, and Modified Wallace multipliers, shown in the comparison in Table 2. The Vedic multiplier (114 LUTs) [4], Array multiplier (116 LUTs) [4], and Booth multiplier (118 LUTs) [4] all require considerably more LUTs than the proposed multiplier, which only requires 68 LUTs. An even greater number of 309 LUTs are used by the Vedic multiplier with the Kogge-Stone adder [17], highlighting the resource efficiency of the proposed design.

Also, the proposed multiplier has a delay of 17.217 ns, which is competitive with other designs like the Reduced Complexity Wallace Multiplier [6] (21.499 ns), although it is greater than some of them like the Vedic multiplier [4] (7.159 ns) and Booth multiplier [4] (6.92 ns). Also, the proposed multiplier uses 1.136 W of power, which is more energy-efficient than the Modified Wallace Multiplier [18], which uses 3.92 W.

The MRPM algorithm claims reduced utilization compared to the design of other conventional multiplier algorithms. MRPM eliminates the production and storage of all partial products characteristic of an array or tree-based multiplier. Rather, it performs iterative value update by elemental shifts and conditional addition. The left shift multiplies one of the operands by two and the right shift divides the other by two, with a possible addition if the operand is odd.This repeated reuse of a small set of logic blocks reduces the need for additional combinational hardware, resulting in significant LUT reduction.

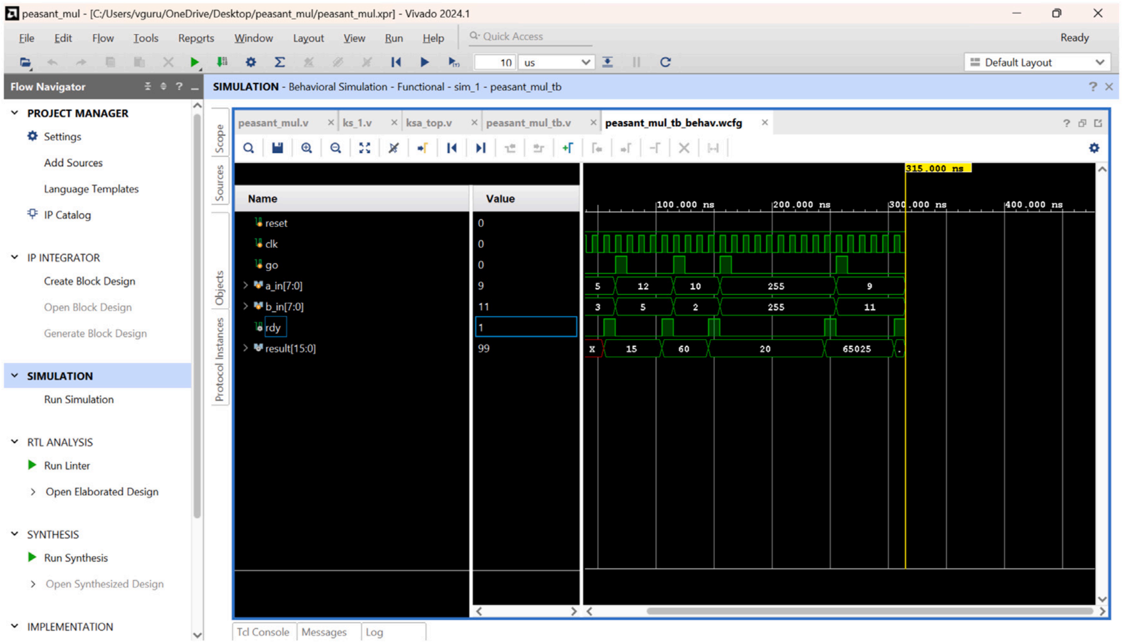

Fig. 5. Simulation result for MRPM algorithm. a_in is the multiplicand, b_in is the multiplier, and the result variable depicts the multiplication product.

Table 2
Post synthesis performance analysis with various 8-bit multipliers.

SL. NO.	Multipliers	No. of bits	No. of LUTs	IOBs	Delay (ns)	Power (W)
1	**Proposed multiplier**	**8**	**68**	**36**	**17.217**	**1.136**
2	Vedic multiplier [4]	8	114	32	7.159	–
3	Array multiplier [4]	8	116	32	9.5	–
4	Booth multiplier [4]	8	118	44	6.92	–
5	Vedic multiplier using Kogge-Stone adder [17]	8	309	–	5.588	–
6	Modified Wallace Multiplier [18]	8	–	–	9.438	3.92
7	Reduced complexity Wallace multiplier using BEC based SORTCSLA [6]	8	224	–	21.499	0.264
8	MSQRTCSLA based MRPM [6]	8	157	–	15.901	0.263
9	Russian multiplier using BEC based SORTCSLA [6]	8	154	–	17	0.765
10	Modified Baugh-Wooley multiplier [19]	8	105	–	5.979	0.59

Although, this flow allows a great reduction in logic utilization, this leads to a significant increase in latency. In order to reduce the latency induced by iterations, a Kogge-Stone Adder (KSA), a fast parallel-prefix adder well suited for large-bit operations is used. The KSA's group propagate and generate mechanism gives the KSA itself compensating faster addition speed at expense of the RPM recursive delay. Area-delay trade-off like the one described here is highly important in biomedical systems, where area and power efficiency are more important than raw speed due to limitations in hardware resources and battery life for wearable and implantable devices.

The proposed multiplier is suitable for biomedical applications [2] due to its unsigned nature, low power consumption, and resource efficiency. These include portable diagnostic tools and wearable health monitors, where unsigned arithmetic is commonly used to analyse signal and sensor data. By ensuring continuous processing, increased device durability, and effective resource utilization, the proposed multiplier satisfies the crucial requirements of these systems.

6. Conclusion

In this work, the presented novel 8-bit unsigned multiplier architecture that integrates Russian Peasant multiplier with Kogge-Stone adder. The motivation for the multiplier design is to reduce utilization and power consumption compared with other traditional multipliers. Through precise design and simulation, this work have demonstrated notable improvements in resource utilization and power consumption while maintaining high computational speed and accuracy which are implemented using the Xilinx Vivado tool 2024.1 and deployed on the Nexys A7 FPGA board. This shows that the multiplier is suitable for applications that need less area and energy consumption particularly in biomedical devices such as wearable health devices and implantable medical devices. The proposed design has better energy efficiency and performance laying the foundation for future advancements in high-performance and low-power biomedical technology. This work emphasizes the importance of innovative algorithms and efficient hardware implementations in the development of the next generation of compact and effective digital systems.

The study focuses on unsigned multiplication, as several steps in biomedical processing such as wearable ECG signal processing devices, usually rely on non-negative data and unsigned arithmetic. However, some operations such as differential signal processing may require some signed support. To address this requirement, the proposed architecture will be extended for signed multiplication in the future works.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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